KV-Cache Scaling

Predicting the context limit before reaching it

Integrated scientific implementation and evidence report | 20 October 2026

Status: FINAL WITH MEASUREMENT LIMITATIONS

Final scope: the user closed additional GPU work on 20 October 2026 to conserve resources. This report delivers the available evidence with explicit measurement limitations; it does not claim every original research objective is experimentally complete.

The analysis, plots and transfer calculations are integrated and reproducible. Verified coverage: 52 primary probes across 9 measured lengths (512-114688 tokens). Requirement status below distinguishes completed measurements from unmet scientific criteria.

Built capability	Evidence / limit
Memory model	Unique tied weights and GQA-aware KV equation; original numerical prediction preserved
GPU observations	52 primary probes across 9 measured lengths (512-114688 tokens)
Recovery	External checkpoint restoration passed after both Colab copies were removed
Analysis	Validated immutable capsules; regressions require multiple lengths; unexplained bytes remain explicit
Transfer	Four config-only model/GPU predictions; conditional published KV-only comparison

The project followed five phases: derivation and preregistration, isolated probes, checkpointing and safety, empirical analysis, and transfer with reproducible presentation. Drive remained disconnected; review provenance is preserved in the original frozen records.

Original experimental limits: Paired prospective held-out validation at L=114688. Actual OOM boundary unobserved; conservative safe-budget censoring is not an OOM threshold.

A separate new T4 allocation produced two successful L=512 replication workers. They match the core-study peaks and cache bytes but are excluded from its fitted curve and repeat counts. See the final appendix and docs/FINAL_SCOPE.md.

Project purpose and deliverable guide

The question is how much context fits before an LLM exhausts usable GPU memory, and why observed peaks differ from an advance equation. We built a predictor and a reproducible experiment: derive first, commit a numerical prediction, measure isolated peaks, and explain the difference.

Required deliverable	Report and repository evidence
1. Derivation with symbols	Section 1; docs/DERIVATION.md; config-derived weights and KV equation.
2. Timestamped advance prediction	Section 2; prereg/formal-2026-10-18-t4/prediction.json and original Git commit.
3. Checkpointing sweep harness	Section 3; src/kv_cache_scaling/sweep.py, probe.py and coordinator_store.py; raw checkpoint capsule.
4. Measured / analytical plots	Sections 4-6; six standalone PNG/PDF figures in artifacts/figures/project-results/.
5. Quantified discrepancy	Section 7; data/processed/project-results/discrepancy.csv; signed components, closure and unaccounted bytes.
6. Cross-model / cross-GPU predictions	Section 8; data/processed/project-results/transfer/; four scenarios and published KV-only verification.

All six output types are available. Their scientific scope differs: the saved curve and KV verification are established; the final hardware OOM bracket and transferred HF context thresholds are not yet validated.

The five phases followed

Phase	Implementation decision
1. Derive and preregister	Read pinned architecture; count tied weights and GQA cache; commit the prediction before model work.
2. Build isolated probes	Use one cold prefill per fresh process, explicit cache/backend/logits policy and synchronized peak counters.
3. Coordinate and preserve	Checkpoint every worker, verify external copies, enforce safe admission and a single OOM allowance.
4. Explain the measurements	Fit inspected KV separately from total peaks; plot both; close signed discrepancy tables with explicit unknowns.
5. Generalize and present	Predict two unrun models on two unrun GPUs; compare published KV figures; provide a CPU-replayable notebook and report.

1. Derivation with defined symbols

Primary model: Qwen/Qwen2.5-1.5B, FP16, batch size 1. Sequence length L is one cold prefill of L tokens with an empty DynamicCache and logits_to_keep=1. HF SDPA is forced to the efficient backend. No generation-to-L, eager fallback or FlashAttention-2 is used.

Single prefill models prompt admission: the full input is processed at once, making transient peaks observable. Generation-to-L would mix a different token-by-token workload and cache-update history into the comparison. We keep it outside the primary contract.

We chose Qwen2.5-1.5B because its roughly 3.09 GB FP16 weights leave room for cache and transient memory on a T4. Its grouped-query attention and tied embeddings also make both architecture-counting pitfalls testable. A larger measured model would leave less room for the phenomenon under study.

Symbol	Meaning	Value
L / B	Input tokens / batch size	variable / 1 measured
P / W	Unique parameters / FP16 weight bytes	1543714304 / 3087428608
U	Usable context-only GPU memory, bytes	15527116800
a / H0	Assumed transient bytes per token / fixed budget reserve	78336 / 536870912
N	Transformer layers	28
H / I	Hidden / intermediate width	1536 / 8960
Q / K / D	Query heads / KV heads / head dimension	12 / 2 / 128
V	Vocabulary	151936
s	Bytes per FP16 cache element	2

q=QD; k=KD. Unique parameters P = VH + N[Hq + 2Hk + qH + q + 2k + 3HI + 2H] + H. The tied input/output embedding is counted once.

W=2P=3087428608 bytes. KV(L,B)=2BLNKDs=28672BL bytes. Using Q in place of K overestimates KV by 6x; duplicating the tied embedding adds 466747392 bytes.

Advance budget model: M(L,B)=W+BL(28672+a)+H0. The central assumption is a=78336 bytes/token and H0=536870912 bytes. L(U,B)=max(0,floor((U-W-H0)/(B(28672+a)))). U is usable memory after context, before weights; context must not be subtracted again.

Original T4 usable budget: 15527116800 bytes. Central estimate: 111232 tokens; assumption range: 76024-131072 after the config cap. This is not a measured confidence interval or an OOM observation. The model-card quality limit is 32768; larger lengths are synthetic memory stress.

P sums embeddings, attention projections and Q/K/V biases, gated MLP weights and RMSNorm scales from the pinned config. M_weights=W; M_kv=28672BL; M_activations is approximated by aBL; M_overhead is budgeted with H0. H0 is an admission reserve, not a measured tensor allocation. One GiB is 1073741824 bytes; one MiB is 1048576 bytes.

2. Original preregistered prediction

Original commit: 7baff5121ed643f27ba06959065855a8913f8e8c. Original Git timestamp: 2026-10-18T19:54:52+05:30. The prediction JSON remains byte-identical to that commit. The later receipt 976158c82ccb10f12817af2ee16b76ade13f0cf3 and updated environment receipts preserve execution provenance without rewriting the advance prediction.

L tokens	KV bytes	Central tensor peak bytes	Central budget bytes
512	14680064	3142216704	3679087616
1024	29360128	3197004800	3733875712
2048	58720256	3306580992	3843451904
4096	117440512	3525733376	4062604288
8192	234881024	3964038144	4500909056

The tensor envelope is W+BL(28672+a). The budget adds H0, so it must not be compared to allocated tensor peaks as though it were another tensor. The central limit is 111232 tokens at U=15527116800 bytes; the assumption range is 76024-131072, capped by config. This sensitivity range is not a confidence interval.

A later fit was frozen after the primary sweep and before L=114688 validation. It is assessed separately below; it never replaces this original prediction or its timestamp.

Hardware and runtime identity

Measured cohort: Tesla T4, GPU-89f781af-1ec0-992b-3dc3-46ef2d0941de; Python 3.11.13, PyTorch 2.6.0+cu124, CUDA runtime 12.4, Transformers 4.51.3, driver 580.82.07. Total GPU bytes: 15637086208; context-only usable bytes: 15527116800. The original preliminary Python 3.13 / torch 2.11 snapshot is not pooled into this cohort.

GPU execution source: ce227af40476676238cbef9126807fd1d42be76b. Model revision: 8faed761d45a263340a0528343f099c05c9a4323. Measurements are grouped by hardware, runtime and protocol identity before analysis.

3. Checkpointed measurement harness

Step	Operational rule
Isolate	A new worker process loads the FP16 model for each point. Measure post-load allocation before any forward pass; compare it with unique-weight bytes.
Measure peak	Synchronize CUDA, reset_peak_memory_stats(), allocate probe inputs/cache, run the prefill and synchronize again. Read max_memory_allocated() and max_memory_reserved(); retain allocated/reserved snapshots and sampled NVML/free-memory telemetry.
Separate lifetimes	Inspect actual cache tensors, then release outputs/cache and measure retained allocation. Fit KV bytes separately; total peak slope also contains transient activations and workspace.
Pin the backend	Force efficient SDPA. Eager attention materializes head-wise L-by-L scores and invalidates the linear transient envelope. FlashAttention-2 is unsupported on this T4 path; no silent fallback is accepted.
Pin cache and allocator	Primary uses empty DynamicCache in a single prefill. Chunked DynamicCache and StaticCache are diagnostics. Set expandable_segments before CUDA initialization in each fresh worker; record version support and effective settings.
Preserve and approach	Write typed results and a fsynced journal after every worker; verify an external checkpoint before continuing. Sweep upward, then use conservative admission and bisection. One deliberate final OOM attempt is allowed; none has been dispatched.

DynamicCache concatenation during chunked updates can create transient copies. StaticCache changes preallocation and, in pinned HF, causal-mask behavior. Neither diagnostic is substituted for the primary contract. Driver peaks from NVML remain sampled, while PyTorch peak counters record allocator high-water marks.

Drive stayed disconnected. The raw checkpoint capsule preserves commands, worker records, package inventories, source fingerprints and coordinator state. Recovery of saved files was tested; restoration of an entire cloud session on the same physical GPU is not claimed.

4. Saved GPU measurements and KV slope

Allocator	L	n	Peak allocated bytes	Peak reserved bytes	Cache bytes
default	512	3	3146681856	3330277376	14680064
default	1024	3	3195448832	3403677696	29360128
default	2048	3	3295079936	3529506816	58720256
default	4096	3	3487002112	3693084672	117440512
default	8192	3	3877137920	4240441344	234881024
default	16384	3	4657409536	5171576832	469762048
default	32768	3	6217952768	6731857920	939524096
default	65536	3	9339039232	10171187200	1879048192
default	114688	3	14020668928	15319695360	3288334336
expandable	512	3	3144715776	3164602368	14680064
expandable	1024	3	3193482752	3202351104	29360128
expandable	2048	3	3291016704	3317694464	58720256
expandable	4096	3	3486084608	3529506816	117440512
expandable	8192	3	3876220416	3925868544	234881024
expandable	16384	3	4656492032	4764729344	469762048
expandable	32768	3	6217035264	6400507904	939524096
expandable	65536	3	9338121728	9695133696	1879048192
expandable	114688	1	14019751424	14644412416	3288334336

Repeated points use fresh worker processes. The cache contains 14680064 bytes at L=512 and reduced logits are FP16 (1,1,151936). Per-length medians and ranges come from the recorded repeats. Kernel traces remain separate diagnostics.

Retained residual fraction: 0.133456% to 0.303312% of retained allocation across primary points.

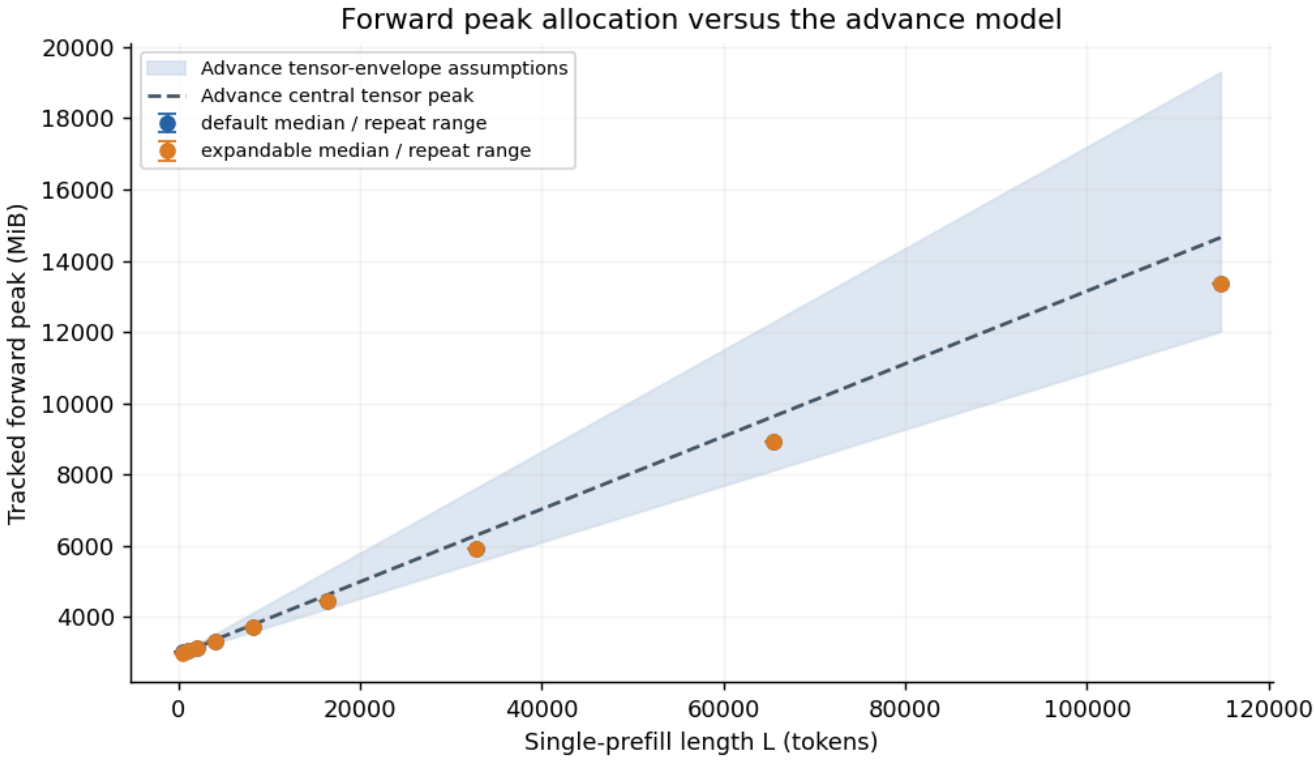
All snapshots respect reserved >= allocated and forward peaks cover synchronized post-forward allocation. Retained-minus-released allocation matches inspected cache bytes. Driver samples are sampled observations, not exact driver high-water marks.

Both allocators cover the five required lengths with three repeats.

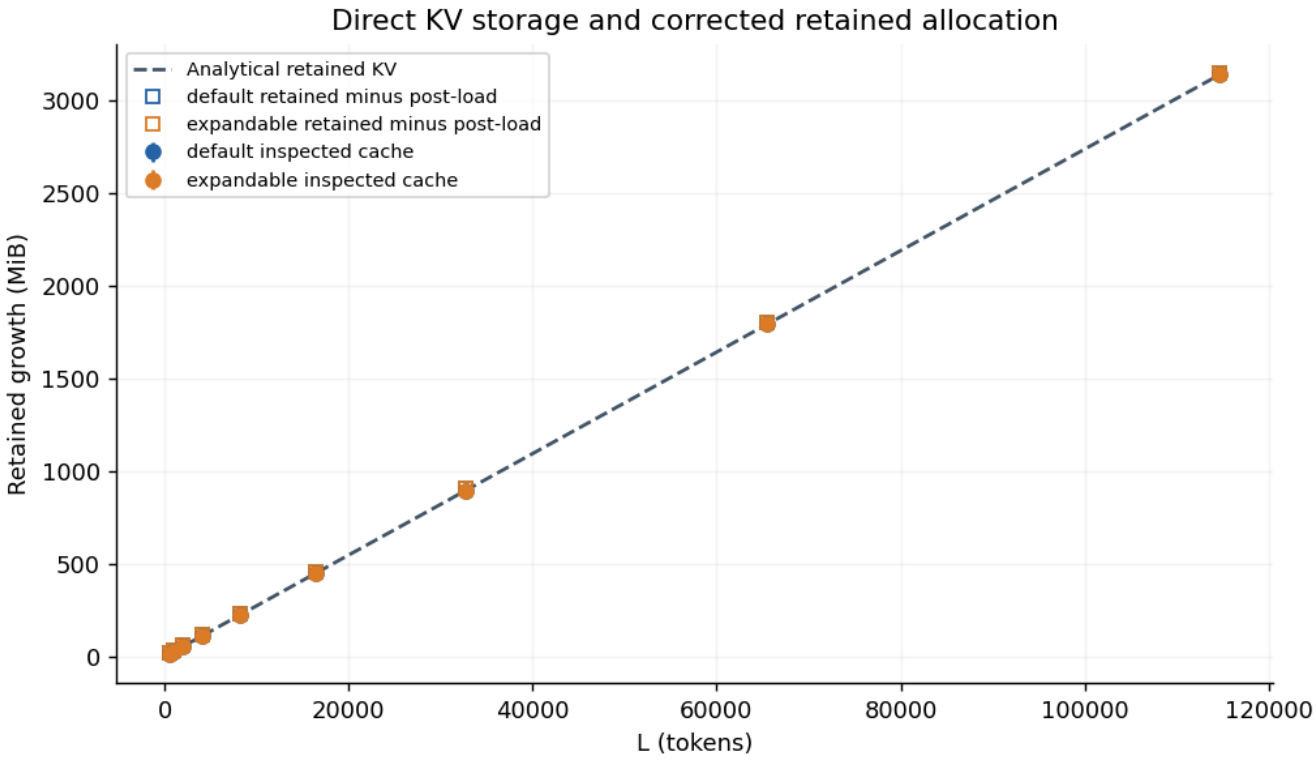
default: directly inspected KV slope = 28672.000000000004 bytes/token; R-squared = 1.0. Total-peak fits remain separate.

expandable: directly inspected KV slope = 28672.000000000004 bytes/token; R-squared = 1.0. Total-peak fits remain separate.

5. Memory curves and retained KV

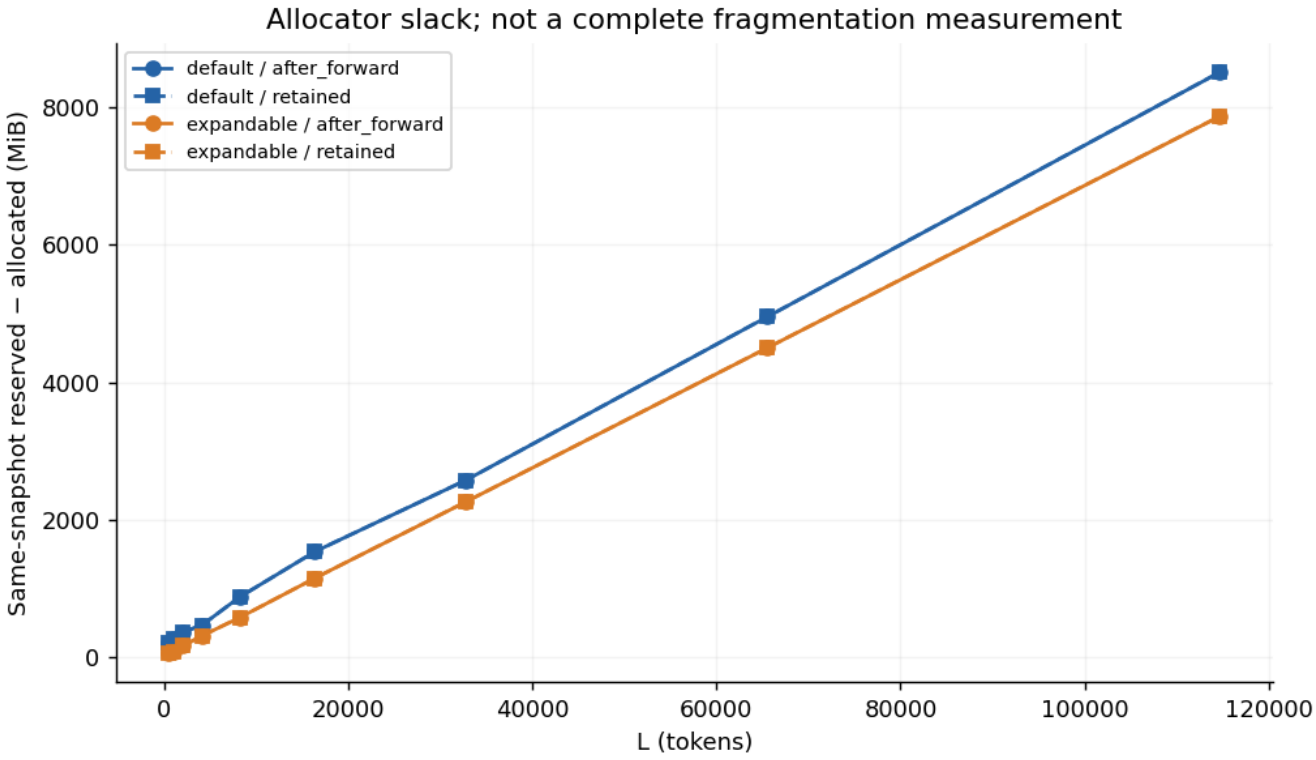


The band is an advance tensor-envelope assumption, not measured uncertainty. Points are observations; the unmeasured curve remains visibly labelled.



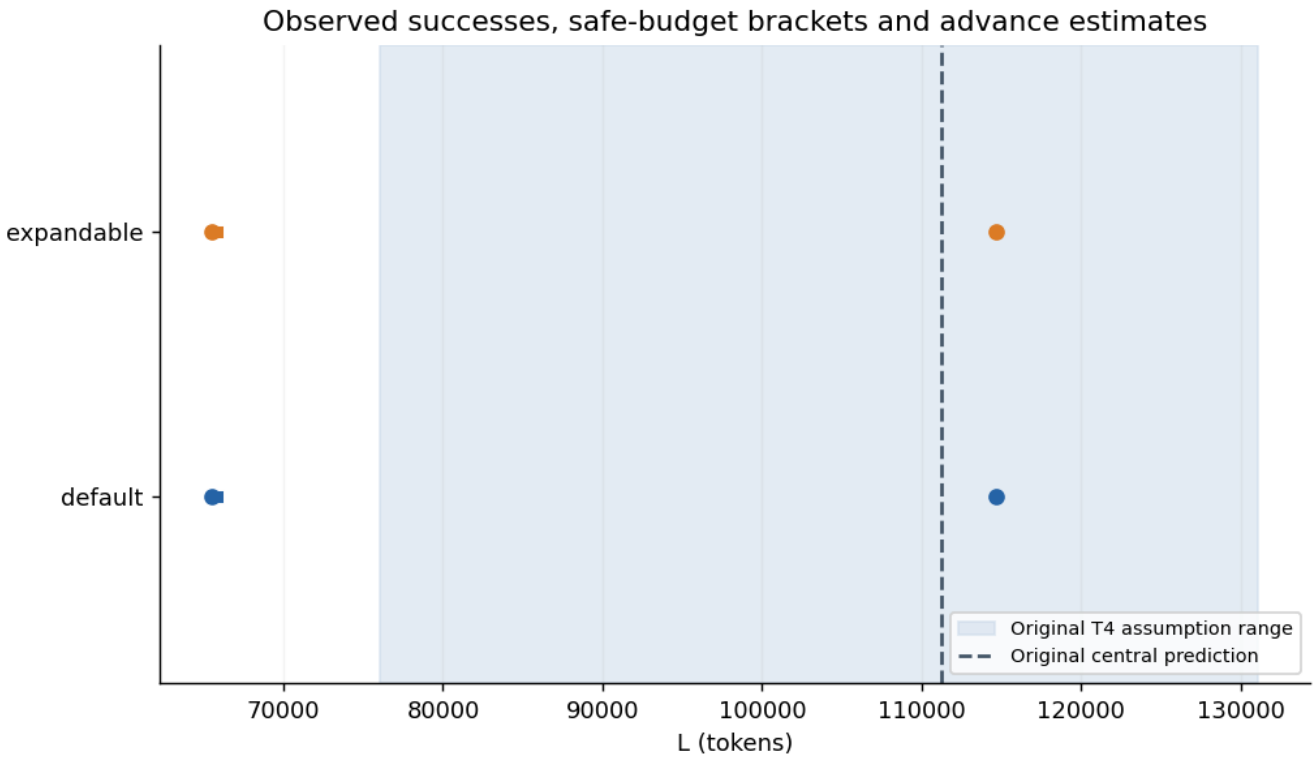
Direct cache and corrected retained allocation are separate quantities. Fits require multiple distinct lengths.

6. Allocator slack and boundary status



Same-snapshot reserved minus allocated is allocator slack. It is not, by itself, fragmentation.

6. Boundary evidence (continued)



Observed success, safe-budget bracket, actual OOM upper bound and advance prediction remain distinct. No hardware threshold is fabricated.

Expandable-minus-default conservative safe-limit interval: -256 to 256 tokens; observed-success delta 0. A common analytical admission ceiling can censor both allocators; this is not a hardware-limit delta.

default: observed success 114688 tokens; OOM upper point None; original prediction error interval None percent. The interval assumes monotone feasibility and does not identify an exact threshold.

expandable: observed success 114688 tokens; OOM upper point None; original prediction error interval None percent. The interval assumes monotone feasibility and does not identify an exact threshold.

Held-out default L=114688, n=3: frozen post-sweep fit predicted 14019203874 bytes; observed median 14020668928; error -0.0104%. This is a later validation, not a replacement advance prediction.

Held-out expandable L=114688, n=1: frozen post-sweep fit predicted 14019751424 bytes; observed median 14019751424; error 0.0000%. This is a later validation, not a replacement advance prediction.

7. Signed prediction discrepancy

Allocator / L / repeats	Median gap bytes	Max closure bytes	Max unaccounted bytes	All repeats
default / 512 (n=3)	4465152	0	0	PASS
default / 1024 (n=3)	-1555968	0	0	PASS
default / 2048 (n=3)	-11501056	0	0	PASS
default / 4096 (n=3)	-38731264	0	0	PASS
default / 8192 (n=3)	-86900224	0	0	PASS
default / 16384 (n=3)	-183238144	0	0	PASS
default / 32768 (n=3)	-375913984	0	0	PASS
default / 65536 (n=3)	-761265664	0	0	PASS
default / 114688 (n=3)	-1339293184	0	0	PASS
expandable / 512 (n=3)	2499072	0	0	PASS
expandable / 1024 (n=3)	-3522048	0	0	PASS
expandable / 2048 (n=3)	-15564288	0	0	PASS
expandable / 4096 (n=3)	-39648768	0	0	PASS
expandable / 8192 (n=3)	-87817728	0	0	PASS
expandable / 16384 (n=3)	-184155648	0	0	PASS
expandable / 32768 (n=3)	-376831488	0	0	PASS
expandable / 65536 (n=3)	-762183168	0	0	PASS
expandable / 114688 (n=1)	-1340210688	0	0	PASS

Summary uses median signed gaps and worst-case absolute closure/unaccounted bytes across repeats. Every repeat-level row is preserved in the accompanying Markdown report and discrepancy CSV.

The signed identity telescopes across post-load, after-forward, retained and released counters. Arithmetic closure is separate from explaining causes. Unknown baseline storage and persistent post-forward growth are retained as unaccounted, even if a plausible source-level hypothesis exists.

For a material gap (at least 10 MiB), supported attribution must reach 90% under both gap and absolute-contribution denominators. Smaller gaps require at most 1 MiB unaccounted and an ill-conditioned-ratio label. Computed coverage status: True.

7. Byte-level discrepancy example

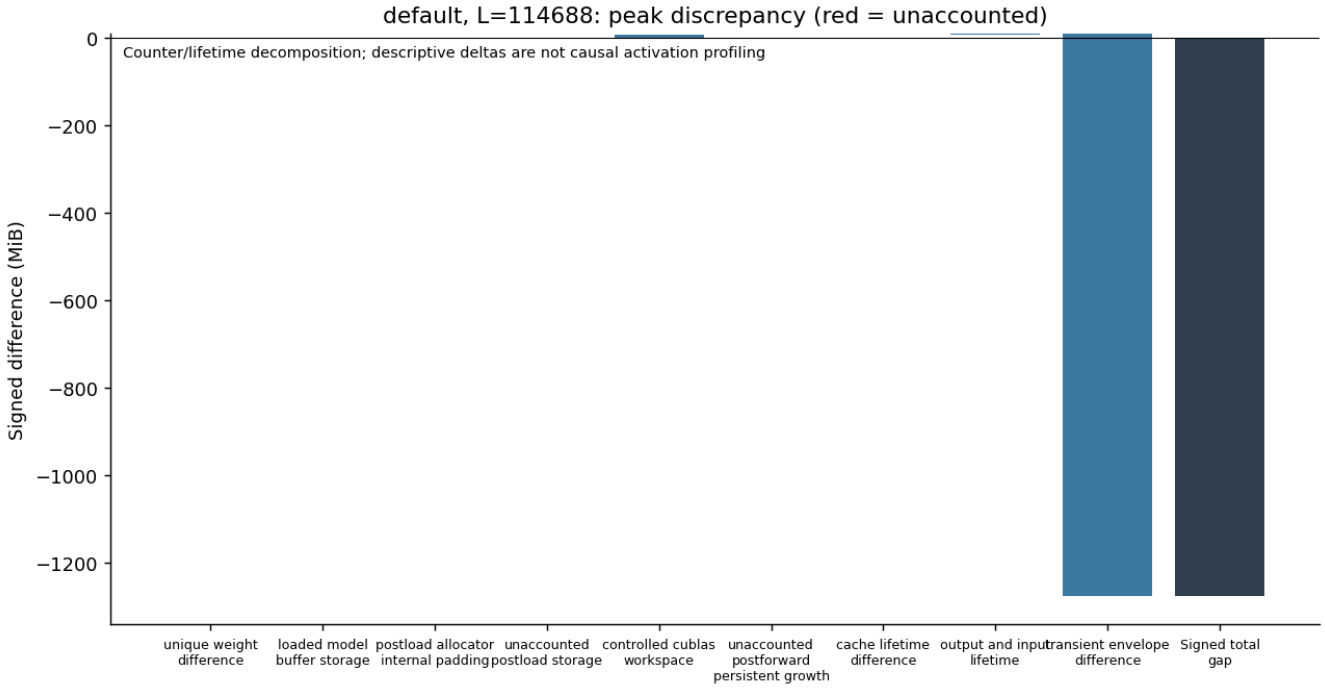
Default allocator, L=114688, repeat=1. Predicted tensor peak: 15359962112 bytes. Measured peak: 14020668928 bytes. Signed difference: -1339293184 bytes. Positive components add memory; negative components reduce the advance envelope.

Component	Signed bytes	Signed MiB	Interpretation
unique weight difference	0	0.000000	Measured storage/counter evidence
loaded model buffer storage	256	0.000244	Measured storage/counter evidence
postload allocator internal padding	917760	0.875244	Measured storage/counter evidence
unaccounted postload storage	0	0.000000	Unaccounted
controlled cublas workspace	8519680	8.125000	Fixed-size inference from L=512 controls
unaccounted postforward persistent growth	0	0.000000	Unaccounted
cache lifetime difference	0	0.000000	Measured storage/counter evidence
output and input lifetime	2139136	2.040039	Measured storage/counter evidence
transient envelope difference	-1350870016	-1288.290039	Descriptive lifetime/counter difference

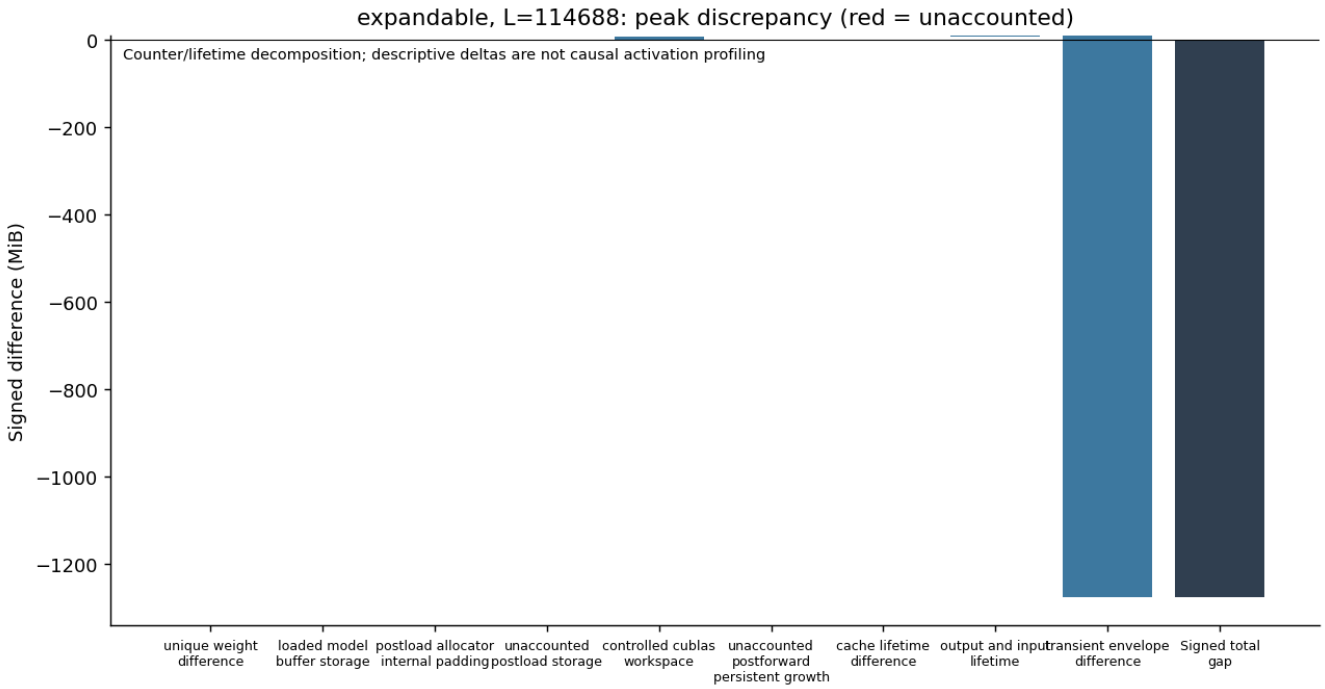
Sum of components: -1339293184 bytes; closure error: 0 bytes; absolute unaccounted: 0 bytes. The separate fixed planning reserve is 536870912 bytes and is not included in this tensor-peak comparison.

The large negative transient-envelope component means the advance tensor envelope was conservative at this length. It is a measured counter/lifetime residual relative to an assumed envelope, not a direct causal measurement of pure activations. Zero unaccounted bytes here therefore establishes this accounting identity, not a complete physical decomposition of allocator fragmentation or an unmeasured OOM gap.

7. Discrepancy components (continued)



Default allocator. Red components remain unaccounted. Transient counter deltas do not establish pure activation memory.



Expandable allocator. The same signed accounting and explanatory-coverage rules apply.

Separate diagnostic comparisons

Mode	Allocator	L	Repeats	Peak delta bytes
chunked_dynamic	default	512	3	1048576
chunked_dynamic	default	2048	3	-100139008
chunked_dynamic	default	8192	3	-465567744
chunked_dynamic	default	32768	3	-1900544000
static_cache	default	512	3	524288
static_cache	default	2048	3	8388608
static_cache	default	8192	3	134217728
workspace_control	default	512	3	-8519680
workspace_control	expandable	512	3	-8519680

Diagnostic minus matching primary median. Chunking and static caching change execution semantics. Workspace interventions change GEMM workspace policy. All are excluded from the primary cache slope and hardware-boundary observations.

Pinned HF 4.51.3 creates an explicit query-by-capacity causal mask for StaticCache. A full-prefill mask is L by L even with efficient SDPA; chunking bounds the query dimension. Peak deltas include this mask behavior, not solely cache storage.

PyTorch 2.6 source allocates 8519680 default cuBLAS workspace bytes on a non-Hopper CUDA GPU. Three matched same-allocator zero-workspace controls directly support L=512 rows. Other lengths use labelled source-supported fixed-size inference only when their persistent delta matches exactly.

8. Transfer to two unrun models and GPUs

Model	GPU	Assumed usable GiB	Raw central L	Capped L	Capped range
Qwen2.5-3B	A100-40GB	37/38/39	251278	32768	32768-32768
Qwen2.5-3B	RTX-4090-24GB	22/23/23.5	132571	32768	32768-32768
Qwen2.5-7B	A100-40GB	37/38/39	109629	109629	72387-131072
Qwen2.5-7B	RTX-4090-24GB	22/23/23.5	39097	39097	22613-55262

Only official config files were downloaded. No 3B/7B checkpoint was loaded and neither target GPU was measured. VRAM budgets and activation envelopes are explicit assumptions. The 3B config cap censors its memory estimate; censoring is not an observed OOM.

Matching KV architecture	Predicted block bytes	Shown MB	Error if MiB	Error if decimal MB
Qwen2.5-3B	589824	0.56	0.4464%	5.3257%
Qwen2.5-7B	917504	0.87	0.5747%	5.4602%

The vLLM team publishes physical 16-token KV blocks across all layers. Base/Instruct KV fields agree in pinned configs. Under a two-byte cache assumption the comparison checks retained KV scaling only. The source labels MB without defining decimal versus binary units, so both displayed-value errors are reported.

The published serving engine has different attention, allocation and scheduling semantics. This evidence does not validate the transferred HF single-prefill peak thresholds, activation envelopes or target usable VRAM.

9. Acceptance status and reproducibility

ID	Status	Criterion
FR1	PASS	Pinned config, KV heads and tied-weight arithmetic
FR2	PASS FOR SAVED ACCEPTANCE	Exact weights/cache verified; residuals reported separately
FR3	PASS FOR SAVED ACCEPTANCE	Original prediction 7baff51 and later receipt 976158c precede model work
FR4	CHECKPOINTED GRID PASS	Fresh workers and per-probe external checkpoint barriers; original acceptance restoration verified
FR5	PASS	Five lengths, three repeats and retained-cache slope required
FR6	SAFE BUDGET CENSORED	One-use OOM bracket and paired safe-budget delta are distinct; an exact hardware-limit delta is not established
FR7	PASS	Closed signed table; unsupported bytes remain unaccounted
FR8	CONDITIONAL KV COMPARISON	Four unrun transfer cells; published KV block comparison with dtype/unit caveats

Regeneration is CPU-only: install the package with requirements-analysis.txt, run kvcache analyze on the immutable capsule, run kvcache generalize, and invoke scripts/build_report.py. Every output directory must be new. Full command lines appear in report.md and the reproduction guide.

Historical acceptance preserved an import-time torchao preflight failure and verified restoration after deleting both Colab checkpoint copies. Removing unused torchao restored cold imports. That bounded acceptance consumed no OOM allowance; full cloud-session restoration of the same GPU identity is not claimed.

The requirement table is computed from saved evidence. Safe-budget boundaries remain distinct from actual hardware OOM thresholds. Unsupported causes remain unaccounted. The original prediction is preserved; the prospective implementation amendment and later receipt record the changed GPU/source before extended probes.

10. Presenting the deliverables

Open notebooks/FINAL_PRESENTATION.ipynb in Colab and Run All. It replays the verified saved evidence on CPU, displays six scientific figures, the derivation, the original prediction and timestamp, harness/checkpointing evidence, discrepancy tables and all four transfer predictions. It also offers individual artifact downloads. No model download, CUDA allocation or Drive mount is needed for this presentation.

Use artifacts/presentation/PROJECT_PRESENTATION.html as the standalone presentation if Colab output scripts fail to load. It contains the same saved-data presentation and embedded graphs. The live-demo guide explains both routes; the decisions and phase guide explains the project in plain language.

What is established, and what remains

The core cohort contains 80 successful workers: 48 primary sweep points, 28 diagnostics and four held-out checks. Both allocators reached 114688 tokens. Two new L=512 replication workers bring the preserved total to 82 across two separate allocations. Further repetitions and the final boundary attempt were deferred by the user on 20 October 2026. No deliberate OOM attempt was dispatched; no hardware OOM upper bound or exact threshold error is claimed.

The discrepancy tables explain fixed-length tracked tensor peak differences under stated accounting assumptions. They do not quantify an unmeasured OOM-threshold gap. The published comparison verifies KV block arithmetic, not the unrun target systems' full context limits. These limitations are part of the deliverable, not hidden exceptions.

Separate 20 October replication

Allocator	L	Repeat	Peak allocated bytes	Peak reserved bytes
default	512	1	3146681856	3330277376
default	512	2	3146681856	3330277376

New allocation: GPU-440a7ac2-62c4-253c-ba99-111c2f1e8466. Both fresh workers match the original default L=512 allocated peak exactly. The coordinator was stopped while waiting in input() after both workers had exited and passed idle checks. No active GPU process remained at closure. The original partial journal was preserved rather than relabelled finished.

Final resource-bounded verification passes for the delivered scope. The original full-experiment verifier remains strict and is not claimed to pass. No more GPU runs are planned under this scope. Transfer thresholds and the exact OOM threshold remain experimental limitations, not hidden completed results.

Primary sources and evidence links

Clean presentation notebook

Five-phase decisions guide

Original prediction commit

Executed audit notebook

Official vLLM KV block table

PyTorch v2.6.0 workspace source

Pinned HF 4.51.3 Qwen2 masking source

Official transfer config URLs/revisions/hashes: configs/transfer/provenance.json. Source capsule hashes, analysis code hashes and report package versions are in provenance and manifest files.